

# NEET Physics — Work, Energy & Power

Ultra Premium Practice Sheet (20 Questions)

**Q1. A force  $F = 3x$  acts on a particle. Work done from  $x=0$  to  $x=2$  m is:**

- A. 6 J
- B. 12 J
- C. 9 J
- D. 3 J

**Q2. Work done by friction on a body moving in a circular path is:**

- A. Zero
- B. Positive
- C. Negative
- D. Infinite

**Q3. Assertion: Work done by centripetal force is zero.**

**Reason: Force is perpendicular to displacement.**

- A. Both correct & explanation
- B. Both correct not explanation
- C. A correct R wrong
- D. Both wrong

**Q4. Kinetic energy depends on:**

- A. Velocity
- B. Mass
- C. Both mass & velocity
- D. None

**Q5. Potential energy of a spring is:**

- A.  $kx$
- B.  $\frac{1}{2} kx^2$
- C.  $kx^2$
- D.  $\frac{1}{2} kx$

**Q6. Work-energy theorem states:**

- A.  $W = \Delta KE$
- B.  $W = KE$
- C.  $W = PE$

D. None

**Q7. Power is rate of:**

- A. Force
- B. Work
- C. Velocity
- D. Energy

**Q8. A 2 kg body moving at 3 m/s stops. Work done:**

- A. 9 J
- B. -9 J
- C. 6 J
- D. -6 J

**Q9. Example of conservative force:**

- A. Friction
- B. Gravity
- C. Tension
- D. Normal

**Q10. Example of non-conservative force:**

- A. Gravity
- B. Electrostatic
- C. Friction
- D. Spring

**Q11. Total mechanical energy in conservative system:**

- A. Increases
- B. Decreases
- C. Constant
- D. Zero

**Q12. Work done in closed path by conservative force:**

- A. Zero
- B. Positive
- C. Negative
- D. Infinite

**Q13. Power =  $F \cdot v$  when angle between  $F$  and  $v$  is:**

- A.  $0^\circ$
- B.  $90^\circ$
- C.  $180^\circ$
- D. None

**Q14. KE equals PE at:**

- A. Maximum velocity
- B. Zero velocity
- C. Intermediate position
- D. Constant velocity

**Q15. Work done by gravity depends on:**

- A. Path
- B. Displacement
- C. Time
- D. Velocity

**Q16. If velocity doubles, KE becomes:**

- A. 2 times
- B. 4 times
- C. Same
- D. Half

**Q17. Unit of power:**

- A. Joule
- B. Watt
- C. Newton
- D. kg

**Q18. Spring force is:**

- A. Constant
- B. Variable
- C. Zero
- D. Infinite

**Q19. Potential energy depends on:**

- A. Position
- B. Velocity
- C. Time

D. Force

**Q20. Efficiency is defined as:**

A. Output/Input

B. Input/Output

C. Work/Time

D. None

## Answer Key

Q1: B  
Q2: C  
Q3: A  
Q4: C  
Q5: B  
Q6: A  
Q7: B  
Q8: B  
Q9: B  
Q10: C  
Q11: C  
Q12: A  
Q13: A  
Q14: C  
Q15: B  
Q16: B  
Q17: B  
Q18: B  
Q19: A  
Q20: A